

Medical Robotics

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Lecture 13

Teleoperation: Two-Port Network Model, Transparency, and Stability

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and integrated reference knowledge.

— **Orange boxes:** comments and clarifications. — **Green boxes:** extended background.
— **Blue boxes:** notes and context. — **Red boxes:** definitions and theorems.

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1. Overview

These notes develop, in a single coherent arc, the complete formal framework for analysing bilateral teleoperation systems:

1. **Two-port network model** — the teleoperator as a cascade of one- and two-port networks; mechanical–electrical analogy; human and environment impedances.
2. **Transparency** — the ideal $Z_t = Z_e$ condition; hybrid matrix formulations; Lawrence’s general four-channel architecture; position-position and position-force schemes; telefunctioning; why perfect transparency requires acceleration measurements.
3. **Passivity** — energy-based stability for nonlinear, uncertain systems; storage function; Lyapunov link; feedback interconnection theorem.
4. **Absolute stability** — Llewellyn’s criterion for LTI two-ports; physical interpretation of the H-matrix entries; the stability index $\eta(\omega)$; why perfect transparency implies $\eta = 1$ (marginal stability).
5. **Wave variables** — passivity under communication delays; Anderson–Spong encoding; scattering matrix.
6. **Example: position-position architecture** — full stability and transparency proofs; sensor requirements; design interpretation of the matching condition.
7. **Exercise: position-force architecture** — stability and transparency analysis.

Order of topics. Stability is logically prior to transparency (a system must be stable before its transparency properties are meaningful). However, transparency is developed first because it can be evaluated through the four-channel architecture before introducing new concept used later to evaluate stability conditions. The fundamental transparency–stability trade-off is formalised in Section 8.2.

2. Two-Port Network Model of a Teleoperator

2.1. Lumped Parameter Networks and Ports

The teleoperation system is modelled as a cascade of **one-port** and **two-port** networks, borrowing the formalism of electrical circuit theory.

- **Lumped parameter element:** a physical entity whose energy (storage elements) or power (dissipative elements) is characterised by a single scalar. Examples: inertia M , viscous damper B , spring K .
- **Network:** a system of lumped parameter elements in series and/or parallel. Examples: RLC circuit; mass-spring-damper system.
- **Port:** a location where energy enters or leaves a network, characterised by conjugate variables: an *effort* $\mathcal{E}$ and a *flow* $\mathcal{F}$, with instantaneous power $P = \mathcal{E} \cdot \mathcal{F}$. Example: the contact point between the human hand and the haptic interface handle.

2.2. Mechanical–Electrical Analogy

Concept	Electrical	Mechanical
Effort	Voltage $V(t)$	Force/Torque $F(t)$
Flow	Current $I(t)$	Linear/angular velocity $\dot{x}(t)$
Resistance	R	Viscous friction B
Inductance	L	Inertia M
Capacitance	$C = 1/K$	Spring compliance $1/K$
Impedance	$Z(s) = V/I$	$Z(s) = F/\dot{x}$ with abuse of notation (mixed time and Laplace domain variables)
Admittance	$Y(s) = 1/Z$	$Y(s) = 1/Z$

Common one-port impedances: $Z_M = Ms$ (inertia; analogous to inductor $Z_L = Ls$), $Z_B = B$ (damper; analogous to resistor $Z_R = R$), $Z_K = K/s$ (spring; analogous to $Z_C = 1/(Cs)$). The factor s in Z_M means that realising a control channel involving inertia requires *differentiating velocity* — a central practical obstacle encountered in Section 5.

2.3. Teleoperator as a Two-Port Network

A **one-port** exchanges energy at a single location. The teleoperator (master + communication channel + slave) forms a **two-port**:

- Port 1 (master side): effort F_m , flow V_m .
- Port 2 (slave side): effort F_s , flow V_s .

The full teleoperation system is a cascade:

$$\underbrace{Z_h}_{\text{1-port: human}} \longleftrightarrow \underbrace{\text{master + channel + slave}}_{\text{2-port teleoperator}} \longleftrightarrow \underbrace{Z_e}_{\text{1-port: environment}}$$

Human and environment impedances: $Z_h(s) = F_h(s)/V_m(s)$, $Z_e(s) = F_e(s)/V_s(s)$. Both are modelled as passive visco-elastic one-ports (mass-spring-damper). The **transmitted impedance** $Z_t = F_m/V_m$ is the impedance that the operator perceives at the master port. Ideal transparency: $Z_t = Z_e$.

2.4. Immittance Representations

Of the four port variables (F_m, V_m, F_s, V_s) , two are independent; the others follow from the system dynamics. Equivalent representations:

- **Impedance $\mathbf{Z}$** : $(F_m, F_s, t) = \mathbf{Z}(V_m, V_s, t)$
- **Admittance $\mathbf{Y}$** : $(V_m, V_s, t) = \mathbf{Y}(F_m, F_s, t)$
- **Hybrid $\mathbf{H}$** : mixes effort and flow variables (two common parameterisations; see Section 3)
- **Scattering $\mathbf{S}$** : encodes wave variables; essential for delay-robust stability (Section 9)

For LTI systems, these operators are matrices of rational transfer functions identifiable from open/short-circuit port measurements.

3. Hybrid Matrix Representations and Transparency

3.1. Two Conventions in the Slides

Comments on notation in the slides. Two different hybrid matrices are both called $\mathbf{H}$ without distinction:

Convention L (Lawrence transparency analysis, slide 34):

$$\begin{bmatrix} F_m \\ V_m \end{bmatrix} = \mathbf{H}_L \begin{bmatrix} V_s \\ -F_s \end{bmatrix}$$

Convention C (circuit h-parameter / Llewellyn form, slides 49–54):

$$\begin{bmatrix} F_m \\ -V_s \end{bmatrix} = \mathbf{H}_C \begin{bmatrix} V_m \\ F_s \end{bmatrix}$$

These define *different matrices*. However, the perfect transparency conditions $H_{11} = H_{22} = 0$, $H_{12} = 1$, $H_{21} = -1$ take the same numerical form in both (verifiable by direct substitution). Convention L is used for the transparency derivation below; Convention C is used for Llewellyn's criterion (Section 8).

3.2. Transparency Analysis via Convention L

Convention L gives $F_m = H_{L,11}V_s - H_{L,12}F_s$ and $V_m = H_{L,21}V_s - H_{L,22}F_s$. Substituting $F_s = Z_e V_s$ (passive environment in contact):

$$Z_t = \frac{F_m}{V_m} = (H_{L,11} - H_{L,12}Z_e)(H_{L,21} - H_{L,22}Z_e)^{-1}$$

Perfect transparency condition. $Z_t = Z_e$ for all Z_e if and only if:

$$H_{L,11} = 0, \quad H_{L,22} = 0, \quad H_{L,12} = 1, \quad H_{L,21} = -1$$

Proof sketch: Setting $Z_t = Z_e$ and requiring the identity to hold for all Z_e , expanding $(H_{L,11} - H_{L,12}Z_e) = Z_e(H_{L,21} - H_{L,22}Z_e)$ and collecting powers of Z_e : coefficient of Z_e^0 gives $H_{L,11} = 0$; coefficient of Z_e^2 gives $H_{L,22} = 0$; coefficient of Z_e^1 gives $H_{L,12} = -H_{L,21}$, satisfying with $H_{L,12} = 1$, $H_{L,21} = -1$.

3.3. Physical Meaning of the H-Matrix Entries

Using Convention C ($F_m = H_{C,11}V_m + H_{C,12}F_s$, $-V_s = H_{C,21}V_m + H_{C,22}F_s$):

Entry	Definition	Physical meaning
$H_{C,11}$	$\left. \frac{F_m}{V_m} \right _{F_s=0}$	Impedance seen at the master port in free motion (slave not in contact, $F_s = 0$). For perfect transparency this must be zero: in free space the operator should feel nothing.
$H_{C,12}$	$\left. \frac{F_m}{F_s} \right _{V_m=0}$	Force transmission coefficient slave→master (master locked). $H_{C,12} = 1$: perfect force feedback. $H_{C,12} \neq 1$: filtered/attenuated.
$H_{C,21}$	$\left. \frac{-V_s}{V_m} \right _{F_s=0}$	Velocity transmission coefficient master→slave in free motion. $H_{C,21} = -1$: slave perfectly tracks master motion.
$H_{C,22}$	$\left. \frac{-V_s}{F_s} \right _{V_m=0}$	Slave admittance (master locked). Velocity developed by the slave in response to a contact force when master is stationary.

Comments on impedance at the master port: $H_{C,11}$ is the impedance at the **master port** ($= F_m/V_m$) when the slave is in free motion ($F_s = 0$). If $H_{C,11} \neq 0$ the operator feels master-side dynamics even with no slave contact — not a leak of the slave impedance, but the master's own inertia and coupling appearing in the free-motion response. This is the “distorted free-motion perception” failure mode.

4. The General Four-Channel Architecture

4.1. Architecture Description (Lawrence, 1993)

Four inter-side channels connect master and slave, in addition to local dynamics on each side:

- C_1 : velocity master→slave (slave tracks master motion — always needed)
- C_2 : force slave→master (haptic force feedback)
- C_3 : force master→slave (force feedforward to slave)
- C_4 : velocity slave→master (position synchronisation from slave side)

The resulting control forces on master and slave are:

$$F_{mc} = -C_m V_m + C_2 F_{s/e} - C_4 V_{s/e} \quad (1)$$

$$F_{sc} = -C_s V_{s/e} + C_1 V_{m/h} + C_3 F_{m/h} \quad (2)$$

4.2. Position-Position Control ($C_2 = C_3 = 0$, $C_1 \neq 0$, $C_4 \neq 0$)

Substituting $C_1 = B_s + K_s/s$ and $C_4 = -(B_m + K_m/s)$ into (1)–(2):

$$F_{mc} = (B_m + K_m/s)(V_e - V_h), \quad F_{sc} = (B_s + K_s/s)(V_h - V_e)$$

Virtual spring-damper interpretation. Both control forces are proportional to the velocity error $V_h - V_e$ (and its integral, i.e. the position error $x_m - x_s$). The system behaves as if master and slave were coupled by a virtual spring K and damper B , each trying to synchronise with the other. The gain of C_1 equals C_s (local slave controller) and $|C_4|$ equals C_m (local master controller) — this symmetry is precisely what enables the absolute stability proof (Section 10).

4.3. Position-Force Control ($C_3 = C_4 = 0$, $C_1 \neq 0$, $C_2 = 1$)

The slave tracks master position (C_1); slave interaction force is fed back to the master at unit gain ($C_2 = 1$). The master controller has no spring term ($C_m = B_m$ only): a restoring spring at the master would distort the perceived environment impedance by superposing a spring-like force on the force reflection signal.

4.4. Parameter Table

Block	Position-Position	Position-Force
Z_m	$M_m s$	$M_m s$
C_m	$B_m + K_m/s$ (PD)	B_m (damping only)
Z_s	$M_s s$	$M_s s$
C_s	$B_s + K_s/s$ (PD)	$B_s + K_s/s$ (PD)
C_1	$B_s + K_s/s$	$B_s + K_s/s$
C_2	0 (none)	1 (unit force feedback)
C_3	0 (none)	0 (none)
C_4	$-(B_m + K_m/s)$	0 (none)

5. Transparency Optimisation and Telefunctioning

5.1. Conditions for Perfect Transparency

Requiring $H_{L,11} = H_{L,22} = 0$, $H_{L,12} = 1$, $H_{L,21} = -1$ in terms of the four-channel parameters gives:

$$C_3 C_2 = I, \quad C_4 = -(C_s + Z_s), \quad C_1 = C_m + Z_m, \quad C_3 = Z_m$$

A **complete four-channel architecture** is required. The channels C_1 and C_4 involve the inertia terms $Z_m = M_m s$ and $Z_s = M_s s$, which require **acceleration measurements**. Accelerometers are susceptible to drift and noise amplification; numerical differentiation of velocity amplifies high-frequency noise. At low frequencies (below ~ 5 Hz, typical of surgical tasks) the inertia terms are negligible and near-transparent operation is achievable with position and velocity measurements alone.

5.2. Telefunctioning

Extension: Telefunctioning (scaled teleoperation). When $C_2 = \alpha \neq 1$, the operator perceives a scaled environment impedance: $Z_t = \alpha Z_e$. The system still provides kinematic similarity but with intentional force scaling.

- $\alpha < 1$: force attenuation; used in **microsurgery** where the slave operates at sub-millimetre scales and tissue forces must be scaled down for the surgeon to retain tactile

sensitivity without overshooting.

- $\alpha > 1$: force amplification; used for handling heavy or distant loads where direct human force is insufficient.

Telefunctioning is distinct from position-only control: the operator still receives a haptic signal, but it represents a dynamically similar (not identical) version of the remote environment. Perfect transparency ($Z_t = Z_e$) is the special case $\alpha = 1$.

6. Passivity: An Energy-Based Stability Tool

6.1. Motivation

Classical signal-based stability tools (Nyquist criterion, root locus) require precise, time-invariant models. Teleoperation systems face uncertain and time-varying human and environment impedances, possible communication delays, and potentially nonlinear dynamics. **Passivity** avoids these difficulties by exploiting the physical impossibility of a passive system generating energy internally.

6.2. Dissipativity and Passivity

Dissipativity (Willems, 1972). For the nonlinear system $\dot{x} = f(x) + g(x)u$, $y = h(x)$ with $f(0) = 0$, $h(0) = 0$, the system is **dissipative** with respect to supply rate $w(u, y) : \mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R}$ if there exists a **storage function** $V(x) \in \mathcal{C}^1 : \mathbb{R}^n \rightarrow \mathbb{R}^+$ such that:

$$V(x(t)) - V(x(t_0)) \leq \int_{t_0}^t w(u(s), y(s)) ds \iff \dot{V}(x) \leq w(u, y)$$

Passivity is dissipativity with supply rate $w(u, y) = y^T u$:

$$V(x(t)) \leq V(x(t_0)) + \int_{t_0}^t y(s)^T u(s) ds$$

A passive system cannot generate internal energy: current energy is at most initial energy plus all energy supplied from outside.

6.3. Link to Lyapunov Stability

If $V(0) = 0$ (origin is a local minimum), V is a Lyapunov candidate and passivity implies:

- $u \equiv 0 \Rightarrow \dot{V} \leq 0$: the origin is an **unforced Lyapunov-stable equilibrium** (passivity $\Rightarrow$ stability of the zero-input system).
- $y \equiv 0 \Rightarrow \dot{V} \leq 0$: the **zero dynamics** is Lyapunov stable.
- **Stabilisation by static output feedback:** $u = -\phi(y)$ with $y^T \phi(y) > 0$ gives $\dot{V} \leq -y^T \phi(y) < 0$ (asymptotic stability). Simplest case: $u = -ky$, $k > 0$.

6.4. Feedback Interconnection of Passive Systems

If two passive systems Σ_1, Σ_2 with storage functions V_1, V_2 are connected in feedback ($u_1 = \pm y_2 + v_1$, $u_2 = \mp y_1 + v_2$), the interconnection is passive with storage function $V_1 + V_2$ and exogenous inputs (v_1, v_2) :

$$\dot{V}_1 + \dot{V}_2 \leq y_1^T u_1 + y_2^T u_2 = y_1^T v_1 + y_2^T v_2$$

This enables modular stability proofs: prove each sub-system (human, teleoperator, environment) passive independently; then their cascade is passive and stable.

6.5. Passivity of the Two-Port Teleoperator

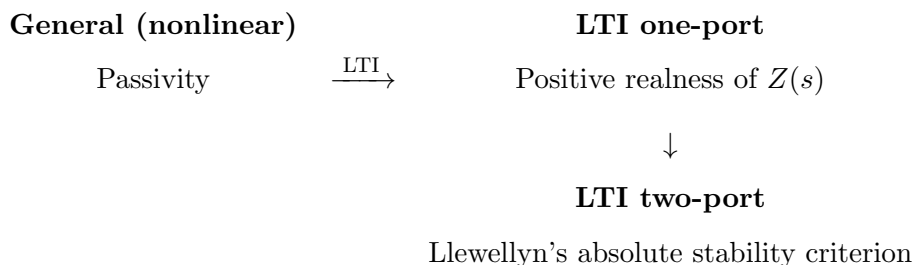
Assuming zero initial energy, the two-port teleoperator is passive if:

$$\int_0^t (F_m(\tau)V_m(\tau) - F_s(\tau)V_s(\tau)) d\tau \geq 0 \quad \forall t > 0$$

$F_m V_m$ is the power flowing *into* the teleoperator from the human; $F_s V_s$ is the power flowing *out* to the environment. Passivity means no net energy is generated inside. This is a *conservative* (sufficient) condition: it must hold for all operator and environment behaviours. A less conservative condition is absolute stability.

7. From Passivity to Absolute Stability: The LTI Stability Chain

For LTI systems the full argument from general passivity to a computable stability criterion proceeds in two steps:



7.1. Positive Realness: Passivity for LTI One-Ports

Positive realness (PR). A rational function $Z(s)$ with real coefficients is **positive-real** if:

1. No poles in the open right-half plane ($\text{Re}[s] > 0$): BIBO stability.
2. Any poles on the imaginary axis are simple with positive residues: purely conservative (not energy-generating) resonances.
3. $\text{Re}[Z(j\omega)] \geq 0$ for all real ω not a pole: the system absorbs (or conserves) energy at every frequency.

An LTI **one-port** is passive if and only if its impedance $Z(s)$ is positive-real.

Extension: geometric interpretation. Condition 3 means $\angle Z(j\omega) \in [-90^\circ, +90^\circ]$ for all ω : the Nyquist plot of $Z(j\omega)$ lies entirely in the closed right half of the complex plane. For a mechanical impedance $Z = Ms + B + K/s$, the real part for $s = j\omega$ is simply B (the damping coefficient): adding damping always improves passivity, while pure inertia and stiffness contribute only imaginary parts and are neutral with respect to passivity.

Positive realness alone is insufficient for **two-port** absolute stability because it only constrains the individual port transfer functions $H_{C,11}$ and $H_{C,22}$, not the cross-port coupling $H_{C,12}H_{C,21}$. An additional condition on the coupling is needed — this is what Llewellyn's criterion provides.

8. Llewellyn's Absolute Stability Criterion

Absolute (unconditional) stability. A linear two-port network is absolutely stable iff

there exists *no* set of passive one-port terminating immittances for which the system is unstable — i.e., it is stable for *any* passive Z_h and Z_e .

Llewellyn's criterion (1952). Using Convention C, a two-port LTI network is **absolutely stable** if and only if:

Condition 1. $H_{C,11}(s)$ and $H_{C,22}(s)$ have no poles in $\text{Re}[s] > 0$; any imaginary-axis poles are simple with positive residues.

Condition 2. For $s = j\omega$, $\forall \omega \geq 0$:

$$\begin{aligned} \text{Re}[H_{C,11}] &\geq 0, & \text{Re}[H_{C,22}] &\geq 0 \\ \eta(\omega) &:= 2 \frac{\text{Re}[H_{C,11}] \text{Re}[H_{C,22}]}{|H_{C,12}H_{C,21}|} - \frac{\text{Re}[H_{C,12}H_{C,21}]}{|H_{C,12}H_{C,21}|} \geq 1 \end{aligned}$$

The criterion is equivalently applicable to any immittance representation.

8.1. Physical Interpretation of the Stability Conditions

Condition 1 requires $H_{C,11}$ and $H_{C,22}$ to be positive-real: each port individually must be passive when the other port is open or short-circuited.

Condition 2 has two parts. The real-part inequalities $\text{Re}[H_{C,11}] \geq 0$ and $\text{Re}[H_{C,22}] \geq 0$ require non-negative energy dissipation at the master and slave ports respectively at every frequency.

The index $\eta(\omega)$ encodes the balance between port dissipation and coupling strength:

- The numerator $2 \text{Re}[H_{C,11}] \text{Re}[H_{C,22}]$ is proportional to the **energy dissipable at the two ports**.
- The denominator $|H_{C,12}H_{C,21}|$ is the **coupling strength**: how much energy can be transferred from one port to the other through the channel.
- $\eta \geq 1$ means: the **coupling energy must not exceed the port dissipation capacity**. If coupling is too strong relative to dissipation, energy circulates and grows.

8.2. Marginal Stability of the Perfectly Transparent Teleoperator

Substituting the perfect transparency conditions ($H_{C,11} = H_{C,22} = 0$, $H_{C,12}H_{C,21} = -1$) into Llewellyn's criterion:

1. No poles: trivially satisfied. ✓
2. $\text{Re}[H_{C,11}] = \text{Re}[H_{C,22}] = 0 \geq 0$. ✓ (equality)
3. $\eta(\omega) = 2 \cdot \frac{0 \cdot 0}{|-1|} - \frac{\text{Re}[-1]}{|-1|} = 0 + 1 = 1$. ✓ (equality)

The stability index equals *exactly* 1 — the boundary of absolute stability. The perfectly transparent teleoperator is **marginally absolutely stable**. Additionally, $H_{C,11} = H_{C,22} = 0$ requires master and slave with zero effective impedance, which is physically unrealisable (all real manipulators have nonzero inertia). The **transparency–stability trade-off is therefore fundamental and unavoidable**: it cannot be circumvented by clever controller design within the two-port framework.

9. Wave Variables and Stability Under Communication Delays (optional)

Comments on transmission delays. Standard bilateral force-velocity control laws, when combined with a communication delay, produce a non-passive channel. The intuitive reason: at the master, the controller computes a force based on a slave velocity that is T seconds old; at the slave, the reverse. This temporal mismatch causes apparent energy injection on each round-trip, destabilising the loop for any $T > 0$ — a failure mode confirmed experimentally before the wave-variable solution was found.

Wave variable approach (Anderson & Spong, 1988). Encoding port variables as wave variables with characteristic impedance $Z_0 > 0$:

$$u_i = \frac{F_i + Z_0 V_i}{\sqrt{2Z_0}}, \quad v_i = \frac{F_i - Z_0 V_i}{\sqrt{2Z_0}}$$

transforms the passivity condition to $\int_0^t (\|u\|^2 - \|v\|^2) d\tau \geq 0$. Transmitting wave variables through a delay T maps $u(t) \mapsto u(t - T)$, which *preserves this inequality for any T* : the wave-variable channel is passive regardless of delay. The associated scattering matrix $\mathbf{S}$ satisfies $\|\mathbf{S}\|_\infty \leq 1$ (a necessary and sufficient passivity condition for LTI two-ports).

The trade-off: the wave encoding introduces Z_0 into the port equations, reducing transparency at high frequencies and causing position drift in free motion. Later extensions (Niemeyer & Slotine, 1991; Hannaford & Ryu, 2002) address these through energy tanks and time-domain passivity control.

10. Example: Position-Position Architecture

10.1. Sensor and Model Requirements

A key practical advantage. Position-position control requires only **joint velocity measurements** (computed from encoder readings by finite differences and low-pass filtering). It requires **no force sensors** and **no dynamic model** of either manipulator: the parameters B_m, K_m, B_s, K_s are *design choices*, not identified physical parameters. This model-free character makes it robust to model uncertainty and straightforward to deploy on heterogeneous hardware. By contrast, the full four-channel architecture for perfect transparency requires the full dynamics model (for channels $C_1 = C_m + Z_m$ and $C_4 = -(C_s + Z_s)$) and accelerometers.

10.2. Hybrid Matrix in Convention C

With $C_2 = C_3 = 0$, solving the closed-loop block diagram gives:

$$\mathbf{H}_C(s) = \begin{bmatrix} Z_m + \frac{C_m Z_s}{Z_s + C_s} & \frac{C_m}{Z_s + C_s} \\ -\frac{C_s}{Z_s + C_s} & \frac{1}{Z_s + C_s} \end{bmatrix}$$

The common denominator of all four entries is $Z_s + C_s = M_s s + B_s + K_s/s$, or equivalently $(M_s s^2 + B_s s + K_s)/s$.

10.3. Absolute Stability Proof

Claim. Position-position is absolutely stable if $M_m, M_s, B_m, B_s, K_m, K_s > 0$ and $K_m/B_m = K_s/B_s$ (i.e., C_m/C_s is a positive real constant).

Step 1 — No RHP poles (Llewellyn Condition 1)

Poles of $H_{C,11}$ and $H_{C,22}$ are zeros of $Z_s + C_s$ with characteristic polynomial:

$$p(s) = M_s s^2 + B_s s + K_s$$

Extension: Routh–Hurwitz criterion.

“Rout criterion” in the transcript → **Routh–Hurwitz criterion** (E. J. Routh, 1877; A. Hurwitz, 1895).

For $p(s) = a_n s^n + \dots + a_0$, a *necessary* condition for all roots to have $\text{Re}[s_i] < 0$ is that all coefficients a_i have the same sign. For $n = 1$ and $n = 2$ this condition is also *sufficient*: for $p = a_2 s^2 + a_1 s + a_0$, all roots satisfy $\text{Re}[s_i] < 0 \Leftrightarrow a_0, a_1, a_2 > 0$. For $n \geq 3$, all-positive coefficients is necessary but not sufficient; the full Routh array must be evaluated.

For $p(s) = M_s s^2 + B_s s + K_s$: $M_s > 0$ (physical inertia), $B_s, K_s > 0$ (design). By Routh–Hurwitz for degree 2, all roots lie strictly in the left-half plane. Condition 1 satisfied. ✓

Step 2 — Non-negative real parts on $s = j\omega$ (Condition 2a)

$$\begin{aligned} \text{Re}[H_{C,22}(j\omega)] &= \frac{B_s}{(\omega M_s - K_s/\omega)^2 + B_s^2} \geq 0 \quad (B_s > 0). \checkmark \\ \text{Re}[H_{C,11}(j\omega)] &= \frac{\omega^2 B_m M_s^2 + M_s(K_m B_s - K_s B_m)}{(\omega M_s - K_s/\omega)^2 + B_s^2} \end{aligned}$$

The numerator has two terms. $\omega^2 B_m M_s^2 \geq 0$ (since $B_m, M_s > 0$). As a **sufficient condition**, we zero the cross-term: $K_m B_s - K_s B_m = 0$, i.e.:

$$\frac{K_m}{B_m} = \frac{K_s}{B_s} \quad \Longleftrightarrow \quad \frac{C_m}{C_s} = \text{const} > 0$$

Under this condition all numerator terms are non-negative. ✓

Design interpretation. $K_m/B_m = K_s/B_s$ requires master and slave PD controllers to have the same damping-to-stiffness ratio. When master and slave are physically similar this is natural; when they differ (e.g., a large haptic master paired with a miniature surgical slave), this constrains achievable gain ratios. Increasing K_m/B_m above K_s/B_s — for instance to improve force perception — makes $\text{Re}[H_{C,11}] < 0$ at some ω , violating absolute stability. This is the precise mechanism linking hardware dissimilarity to the stability margin.

Step 3 — Stability index $\eta(\omega) \geq 1$ (Condition 2b)

Under the matching condition:

$$\eta(\omega) = 2 \frac{\text{Re}[H_{C,11}] \text{Re}[H_{C,22}]}{|H_{C,12} H_{C,21}|} - \frac{\text{Re}[H_{C,12} H_{C,21}]}{|H_{C,12} H_{C,21}|} = 1 \quad \forall \omega \geq 0 \checkmark$$

(The computation involves substituting the matched-gain expression; details in Lawrence, 1993.)

Conclusion: all three Llewellyn conditions are satisfied. The position-position architecture is **absolutely stable** under the sufficient conditions stated. □

$\eta = 1$: the explanation in a sentence. Position-position achieves absolute stability *because* it uses no force channels ($C_2 = C_3 = 0$). Without force channels, $|H_{C,12} H_{C,21}|$ is minimised; the coupling is small enough relative to port dissipation to satisfy $\eta = 1$. Adding

any force channel ($C_2 \neq 0$) increases $|H_{C,12}H_{C,21}|$, pushing η below 1 — the stability-transparency trade-off in concrete form.

10.4. Transparency Analysis

Claim. Position-position is not perfectly transparent, for two reasons.

Entry	Value	Required	Transparent?
$H_{C,11}$	$Z_m + C_m Z_s / (Z_s + C_s) \neq 0$	0	No
$H_{C,12}$	$C_m / (Z_s + C_s) \neq 1$	1	No
$H_{C,21}$	$-C_s / (Z_s + C_s) \neq -1$	-1	No
$H_{C,22}$	$1 / (Z_s + C_s) \neq 0$	0	No

Non-ideal force tracking ($H_{C,12} \neq 1$). The force transmission $C_m / (Z_s + C_s)$ is a dynamic filter. At high frequencies it tends to 0: impulsive contact forces are attenuated and the operator cannot feel fast transients. At low frequencies it approaches $K_m / K_s \neq 1$ in general. Since $C_2 = 0$ there is no direct force channel; slave-environment forces reach the master only indirectly through the virtual coupling.

Distorted free-motion perception ($H_{C,11} \neq 0$). The operator perceives $H_{C,11} \neq 0$ as an impedance even when the slave is not touching anything ($F_s = 0$). The master inertia and the virtual spring-damper generate a restoring force at the master proportional to its velocity. In surgical practice, this prevents the surgeon from detecting the moment of first tissue contact, since free motion and very gentle contact produce qualitatively similar sensations at the master.

11. Exercise: Position-Force Architecture

Analyse the position-force scheme ($C_2 = 1$, $C_3 = C_4 = 0$, $C_m = B_m$): is it absolutely stable? is it transparent?

Solution outline. Generated by AI and NOT verified, check is left as an exercise.

Hybrid matrix. With $C_2 = 1$, $C_3 = C_4 = 0$, closed-loop analysis gives:

$$\mathbf{H}_C^{pf}(s) = \begin{bmatrix} Z_m + \frac{C_m(Z_s + 1)}{Z_s + C_s} & \frac{C_m + 1}{Z_s + C_s} \\ -\frac{C_s}{Z_s + C_s} & \frac{1}{Z_s + C_s} \end{bmatrix}$$

(The “+1” in $H_{C,12}^{pf}$ arises from the $C_2 = 1$ direct force path to the master.)

Absolute stability. Condition 1 (no RHP poles) holds: denominator $Z_s + C_s$ is unchanged. However, the coupling product is now:

$$|H_{C,12}^{pf} H_{C,21}^{pf}| = \frac{|C_m + 1| |C_s|}{|Z_s + C_s|^2} > \frac{|C_m| |C_s|}{|Z_s + C_s|^2} = |H_{C,12}^{pp} H_{C,21}^{pp}|$$

The denominator of $\eta(\omega)$ is larger. Unless the numerator also increases proportionally (which it does not, since $H_{C,11}$ grows more slowly), $\eta(\omega) < 1$ at some frequencies. **Position-force is potentially unstable:** there exist passive environments and operators for which the closed loop is unstable. Stable operation is possible for specific parameter ranges but is not guaranteed across all passive terminations.

Transparency. $H_{C,12}^{pf} = (C_m + 1)/(Z_s + C_s) \neq 1$ in general, and $H_{C,11}^{pf} \neq 0$. Position-force is also **not perfectly transparent**, but provides strictly better force feedback than position-position (the direct $C_2 = 1$ channel gives a less attenuated force transmission).

Summary. Position-force occupies a different point on the stability-transparency trade-off curve from position-position: better haptic fidelity, weaker stability guarantee. Position-position (maximally stable, minimal transparency) and the full four-channel architecture (marginally stable, maximal transparency) form the two extremes.

12. Summary and Key Conceptual Threads

- **The two-port model unifies circuit and mechanical analysis.** The force-velocity / voltage-current analogy allows circuit-theory tools (hybrid matrices, Llewellyn's criterion) to be applied directly to bilateral teleoperation. The factor s in $Z_M = Ms$ is the root of the acceleration-measurement problem.
- **Two hybrid conventions coexist in the slides.** Convention L (Lawrence) for transparency; Convention C (h-parameter / Llewellyn) for absolute stability. The perfect transparency conditions take the same numerical form in both.
- **Perfect transparency requires a complete four-channel architecture with acceleration measurements.** Position-position and position-force both sacrifice transparency for simplicity and stability.
- **Telefunctioning** generalises transparency to scaled teleoperation ($Z_t = \alpha Z_e$): the standard operating mode for microsurgery.
- **Passivity connects energy physics to Lyapunov stability.** The storage function is a Lyapunov candidate; the feedback interconnection theorem enables modular proofs; passivity applies to uncertain, nonlinear, time-varying systems.
- **The LTI stability chain is: passivity $\rightarrow$ positive realness $\rightarrow$ Llewellyn.** For LTI one-ports, passivity $\Leftrightarrow$ PR. For LTI two-ports, absolute stability $\Leftrightarrow$ Llewellyn's criterion (two conditions: port positive-realness and $\eta(\omega) \geq 1$).
- **Perfect transparency forces $\eta = 1$ (marginal stability).** A formally proved, unavoidable consequence of the network structure.
- **Communication delays do not destroy passivity by themselves;** standard bilateral control laws do. Wave variables (Anderson & Spong, 1988) restore passivity for any delay magnitude by exploiting the lossless character of a delay in wave coordinates.
- **Position-position is absolutely stable ($\eta = 1$, sufficient conditions: matched PD gains) but not transparent ($H_{C,11} \neq 0$: distorted free-motion; $H_{C,12} \neq 1$: non-ideal force tracking).** It requires only velocity measurements and no robot model. Adding force feedback ($C_2 \neq 0$) improves transparency but reduces η below 1, risking instability for some passive terminations.

Key topics for revision (midterm quiz scope).

1. Mechanical–electrical analogy; why $Z_M = Ms$ implies acceleration measurement.
2. Two-port teleoperator structure; human/environment as one-ports.
3. Two hybrid matrix conventions; when each is used.
4. Derivation of Z_t ; perfect transparency conditions (with proof).

5. Four-channel architecture: role of each C_i ; parameter tables for both architectures; virtual spring-damper interpretation.
6. Telefunctioning: definition; role of C_2 ; microsurgery context.
7. Dissipativity and passivity: storage function, supply rate, the inequality; physical meaning.
8. Lyapunov link: unforced stability, zero dynamics, output feedback stabilisation.
9. Feedback interconnection theorem.
10. The LTI stability chain: positive realness (three conditions; geometric interpretation); Llewellyn's criterion (two conditions; physical meaning of η).
11. Routh–Hurwitz criterion: statement for general n ; necessary and sufficient for $n = 2$.
12. Why $\eta = 1$ for perfect transparency (algebraic proof).
13. Communication delays: why they cause instability; wave variables; scattering matrix.
14. Position-position stability proof: denominator polynomial (Routh), $\text{Re}[H_{C,11}]$, $\text{Re}[H_{C,22}]$, matching condition $K_m/B_m = K_s/B_s$, $\eta = 1$.
15. Position-position transparency failure: $H_{C,12} \neq 1$ and $H_{C,11} \neq 0$; surgical significance of each.
16. Position-force: why it is potentially unstable; why it is not perfectly transparent; its place on the stability-transparency trade-off.

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